

1. Let n_L be the number of layers in the network
2. Let n_P be the number of pos
3. Let n_S be the sequence length

Let KC be a 3D matrix with dimensions n_L, n_P, n_S such that

1. $KC[x]$ is a 2D matrix of dimensions n_P, n_S
2. $KC[x, y]$ is a vector of length n_S
3. $KC[x, y, z]$ is a scalar

Let KC be stored as a linear array at address mkC

1. the address of $mkC[x]$ is $mkC + x \times (n_P \times n_S)$
2. the address of $mkC[x, y]$ is $mkC + x \times (x \times n_P \times n_S) + (y \times n_S)$
3. the address of $mkC[x, y, z]$ is $mkC + x \times (x \times n_P \times n_S) + (y \times n_S) + z$

1 Instructions to LLM

1. Write an optimized C function that computes $\sum KC[x, y]$.
2. Return a `.c` file — the definition
3. Return a `.h` file — the declarationdefinition

Return